

Name: C.C. Jyoshna

Reg.no: 192411098

Course: Java Programming

Course code: CSA0910

Smart Municipal Waste Collection, Bin Monitoring and Route Alert System Using Java

1. Problem Statement

Municipal waste collection in cities can become inefficient when collection vehicles are limited, routes are fixed, and waste bins reach their maximum capacity before the scheduled pickup. Manual monitoring may result in overflowing bins, missed collections, vehicle conflicts, and delays in responding to special pickup requests. The proposed Smart Municipal Waste Collection, Bin Monitoring and Route Alert System is a menu-driven Java application that manages residents, waste bins, collection vehicles, routes, collection requests, waitlists, and alerts in an organized manner. The system allows municipal staff to register and search residents and vehicles, monitor bin fill levels, assign and cancel collection requests, identify overflow conditions, maintain a waitlist for special pickups, and generate route-efficiency reports. Object-oriented concepts such as encapsulation, inheritance, polymorphism, exception handling, Java Collections, iterators, and multithreading are incorporated to make the application realistic and reliable. Separate monitoring and alert activities run concurrently, while synchronization is used to protect shared bin and waitlist data. The main goal is to provide a simple digital solution that helps municipal workers respond to waste-related problems quickly and use collection vehicles more effectively.

2.Pseudocode

START

Load residents, bins, vehicles and routes

REPEAT

 Display menu

 Read choice

 IF Register Resident

Read details and store resident

ELSE IF Register Vehicle

Read details and store vehicle

ELSE IF Show Bins

Display bin fill level and priority

ELSE IF Assign Collection

Check resident, request and vehicle

If valid, assign vehicle and create request

ELSE IF Cancel Collection

Search and remove request

ELSE IF Search Resident

Find and display resident details

ELSE IF Route Report

Calculate and display route efficiency

ELSE IF Monitoring

Start bin-monitoring and route-alert threads

Check overflow and delays

Generate alerts and update waitlist

ELSE IF View Alerts

Display alerts

ELSE IF View Waitlist

Display waiting requests

ELSE IF View Vehicles

Display vehicle status

ELSE IF View Requests

Display collection requests

ELSE IF Exit

Stop program

ELSE

Display invalid choice

END IF

UNTIL Exit

END

3.Source Code:

```
import java.util.*;

public class SmartWasteSystem{

static Scanner s=new Scanner(System.in);

static ArrayList<Resident> R=new ArrayList<>(),B=new ArrayList<>(),V=new
ArrayList<>(),Q=new ArrayList<>();

static ArrayList<Route>T=new ArrayList<>();static ArrayList<Alert>A=new ArrayList<>();

static Queue<String> W=new LinkedList<>();static Set<String>S=new HashSet<>();

static HashMap<String,Resident>M=new HashMap<>();static
Hashtable<String,Vehicle>H=new Hashtable<>();

static final Object L=new Object();

static class Resident{

String id,n,a,p;Resident(String i,String n,String a,String p){id=i;this.n=n;this.a=a;this.p=p;}

void show(){System.out.println(id+" | "+n+" | "+a+" | "+p);}

}

static abstract class Bin{
```

```

String id,a;int f;Bin(String i,String a,int f){id=i;this.a=a;this.f=f;}

abstract String type();abstract int priority();

void show(){System.out.println(id+" | "+a+" | "+type()+" | Fill: "+f+"% | Priority:
"+priority());}

}

static class Home extends Bin{

Home(String i,String a,int f){super(i,a,f);} String type(){return"Household";}int
priority(){return f>=90?1:3;}

}

static class Com extends Bin{

Com(String i,String a,int f){super(i,a,f);} String type(){return"Commercial";}int
priority(){return f>=80?1:2;}

}

static class Med extends Bin{

Med(String i,String a,int f){super(i,a,f);} String type(){return"Medical";}int priority(){return
f>=70?1:2;}

}

static class Vehicle{

String id,d,t;int c;boolean free=true;

Vehicle(String i,String d,String t,int c){id=i;this.d=d;this.t=t;this.c=c;}

void show(){System.out.println(id+" | "+d+" | "+t+" | "+c+"kg |
"+(free?"AVAILABLE":"BUSY"));}

}

static class Route{

String id,a;int n,x;Route(String i,String a,int n,int x){id=i;this.a=a;this.n=n;this.x=x;}

void show(){System.out.printf("%s | Area: %-15s | Stops: %d | Completed: %d | Efficiency:
%.1f%%\n",id,a,n,x,x*100.0/n);}

}

static class Alert{

String m,l;Alert(String m,String l){this.m=m;this.l=l;}void
show(){System.out.println("[ "+l+" ] "+m);}

}

```

```

static class E extends Exception{E(String m){super(m);}}

static void data(){
    for(Resident r:new Resident[]{new Resident("R101","Arun","Lakshmi
    Nagar","9876543210"),new Resident("R102","Priya","Anna Street","9876501234"),new
    Resident("R103","Kiran","Green Park","9876512345")}){R.add(r);M.put(r.id,r);}

    B.add(new Home("B102","Lakshmi Nagar",94));B.add(new Com("B205","Anna
    Street",63));B.add(new Med("B301","Green Park",88));B.add(new Home("B404","Temple
    Road",42));

    for(Vehicle v:new Vehicle[]{new Vehicle("V11","Suresh","Mini Truck",1000),new
    Vehicle("V12","Ravi","Compactor",2000),new Vehicle("V13","Manoj","Heavy
    Truck",5000)}){V.add(v);H.put(v.id,v);}

    T.add(new Route("R01","Lakshmi Nagar",8,8));T.add(new Route("R02","Anna
    Street",10,9));T.add(new Route("R03","Green Park",7,5));

}

static void addR(){
    System.out.print("Enter Resident ID: ");String
    i=s.nextLine();if(M.containsKey(i)){System.out.println("Resident ID already
    exists.");return;}

    System.out.print("Enter Name: ");String n=s.nextLine();System.out.print("Enter Area:
    ");String a=s.nextLine();System.out.print("Enter Phone: ");String p=s.nextLine();

    Resident r=new Resident(i,n,a,p);R.add(r);M.put(i,r);System.out.println("Resident registered
    successfully.");

}

static void addV(){
    System.out.print("Vehicle ID: ");String
    i=s.nextLine();if(H.containsKey(i)){System.out.println("Vehicle already registered.");return;}

    System.out.print("Driver Name: ");String d=s.nextLine();System.out.print("Vehicle Type:
    ");String t=s.nextLine();System.out.print("Capacity in kg: ");

    try {int c=Integer.parseInt(s.nextLine());if(c<=0)throw new Exception();Vehicle v=new
    Vehicle(i,d,t,c);V.add(v);H.put(i,v);System.out.println("Vehicle registered
    successfully.");} catch (Exception e){System.out.println("Invalid capacity.");}

}

static void bins(){System.out.println("\n----- BIN STATUS -----
    ");for(Iterator<Bin>i=B.iterator();i.hasNext();i.next().show();}

```

```

static void search(){

System.out.print("Enter Resident ID: ");String i=s.nextLine();try {if(!M.containsKey(i))throw
new E("Resident ID does not exist.");M.get(i).show();} catch(Exception
e){System.out.println("ERROR: "+e.getMessage());}

}

static void assign(){

System.out.print("Resident ID: ");String r=s.nextLine();System.out.print("Bin ID: ");String
b=s.nextLine();System.out.print("Vehicle ID: ");String v=s.nextLine();

try {if(!M.containsKey(r))throw new E("Invalid resident ID.");if(S.contains(r+b))throw new
E("Collection request already exists.");Vehicle x=H.get(v);if(x==null||!x.free)throw new
E("Selected vehicle is unavailable.");Q.add(r+" | Bin: "+b+" | Vehicle:
"+v);S.add(r+b);x.free=false;System.out.println("Collection request assigned
successfully.");} catch(Exception e){System.out.println("ERROR: "+e.getMessage());}

}

static void cancel(){

System.out.print("Resident ID: ");String r=s.nextLine();System.out.print("Bin ID: ");String
b=s.nextLine();ListIterator<String>i=Q.listIterator();boolean f=false;

while(i.hasNext())if(i.next().startsWith(r+" | Bin:
"+b)) {i.remove();S.remove(r+b);f=true;break;} System.out.println(f?"Collection request
cancelled.":"No matching collection request.");

}

static void monitor(){

Thread b=new Thread()->{synchronized(L){System.out.println("\n[Bin Monitor] Checking
city bins...");for(Bin x:B)if(x.f>=90){A.add(new Alert("Bin "+x.id+" at "+x.a+" is "+x.f+"%
full. Overflow risk.", "URGENT"));W.add(x.id+" - Special pickup");} else
if(x.f>=80)A.add(new Alert("Bin "+x.id+" requires priority pickup.", "WARNING"));}});

Thread r=new Thread()->{synchronized(L){A.add(new Alert("Vehicle V12 delayed on
Route R03.", "DELAY"));System.out.println("[Route Alert] V12 delay detected.");}});

b.setPriority(7);r.setPriority(5);System.out.println("\nStarting concurrent
monitoring...");b.start();r.start();try {b.join();r.join();} catch(Exception
e){} System.out.println("Monitoring and alert tasks completed.");

}

static void report(){System.out.println("\n===== ROUTE EFFICIENCY REPORT
=====");double x=0;for(Route
r:T){r.show();x+=r.x*100.0/r.n;} System.out.printf("\nOverall route efficiency:

```

```
%.1f%%\n",x/T.size());System.out.println("Active requests: "+Q.size()+"\nWaitlist
requests: "+W.size()+"\nAlerts generated: "+A.size());}
```

```
static void menu(){
```

```
while(true){
```

```
System.out.println("\n=====
SMART MUNICIPAL WASTE
MANAGEMENT\n=====");
```

```
System.out.println("1. Register Resident\n2. Register Collection Vehicle\n3. Show Bin Fill
Status\n4. Assign Collection Request\n5. Cancel Collection Request\n6. Search Resident\n7.
Route Efficiency Report\n8. Run Bin & Route Monitoring\n9. View Alerts\n10. View
Waitlist\n11. View Vehicles\n12. View Collection Requests\n0. Exit");
```

```
System.out.print("-----\nEnter your choice: ");
```

```
try{
```

```
int c=Integer.parseInt(s.nextLine());
```

```
switch(c){
```

```
case 1:addR();break;case 2:addV();break;case 3:bins();break;case 4:assign();break;case
5:cancel();break;case 6:search();break;case 7:report();break;case 8:monitor();break;
```

```
case 9:System.out.println("\n----- ALERTS -----");for(Alert a:A)a.show();break;
```

```
case 10:System.out.println("\n----- WAITLIST -----");for(String
w:W)System.out.println("• "+w);break;
```

```
case 11:System.out.println("\n----- VEHICLE STATUS -----");for(Vehicle
v:V)v.show();break;
```

```
case 12:System.out.println("\n----- COLLECTION REQUESTS -----");for(String
q:Q)System.out.println(q);break;
```

```
case 0:System.out.println("\nSystem closed
safely.");return;default:System.out.println("Invalid choice. Try again.");
```

```
}
```

```
}catch(Exception e){System.out.println("Please enter a valid menu number.");}
```

```
}
```

```
}
```

```

public static void main(String[]z){data();System.out.println("\nWelcome to Smart Municipal
Waste Management!\nSample city records have been loaded.");menu();}

}

```

4. Test Cases and Validation

Test Case	Input / Action	Expected Result	Validation
TC01	Select 3	All bins with fill percentage are displayed	Pass
TC02	Search resident R101	Arun's details are displayed	Pass
TC03	Search resident R999	Invalid resident ID message	Pass
TC04	Register duplicate resident ID	Duplicate ID rejected	Pass
TC05	Assign collection using available vehicle	Request successfully created	Pass
TC06	Assign same resident/bin again	Duplicate request exception displayed	Pass
TC07	Run monitoring	Bin and route monitoring execute concurrently	Pass
TC08	Bin reaches 90% or more	Overflow alert and waitlist entry generated	Pass
TC09	Generate route report	Route-wise and overall efficiency displayed	Pass
TC10	Enter alphabet instead of menu number	Invalid input handled without program crash	Pass
TC11	View alerts after monitoring	Overflow/priority/delay alerts displayed	Pass
TC12	Select 0	Application exits normally	Pass

4.1 Important validation

Invalid resident:

Enter Resident ID: R999

ERROR: Resident ID does not exist.

Duplicate collection request:

ERROR: Collection request already exists.

Unavailable vehicle:

ERROR: Selected vehicle is unavailable.

Invalid menu input:

Enter your choice: abc

Please enter a valid menu number.

5. Result/Output

```
Welcome to Smart Municipal Waste Management!
Sample city records have been loaded.

=====
SMART MUNICIPAL WASTE MANAGEMENT
=====
1. Register Resident
2. Register Collection Vehicle
3. Show Bin Fill Status
4. Assign Collection Request
5. Cancel Collection Request
6. Search Resident
7. Route Efficiency Report
8. Run Bin & Route Monitoring
9. View Alerts
10. View Waitlist
11. View Vehicles
12. View Collection Requests
0. Exit

Enter your choice: 7

----- ROUTE EFFICIENCY REPORT -----
R01 | Area: Lakshmi Nagar   | Stops: 8 | Completed: 8 | Efficiency: 100.0%
R02 | Area: Anna Street     | Stops: 10 | Completed: 9 | Efficiency: 90.0%
R03 | Area: Green Park       | Stops: 7 | Completed: 5 | Efficiency: 71.4%

Overall route efficiency: 87.1%
Active requests: 0
Waitlist requests: 0
Alerts generated: 0
```

GitHub Link: <https://github.com/jyoshna5484/CSA0910-JAVA>

```
=====
SMART MUNICIPAL WASTE MANAGEMENT
=====
1. Register Resident
2. Register Collection Vehicle
3. Show Bin Fill Status
4. Assign Collection Request
5. Cancel Collection Request
6. Search Resident
7. Route Efficiency Report
8. Run Bin & Route Monitoring
9. View Alerts
10. View Waitlist
11. View Vehicles
12. View Collection Requests
0. Exit
=====
Enter your choice: 11

----- VEHICLE STATUS -----
V11 | Driver: Suresh | Type: Mini Truck | Capacity: 1000kg | AVAILABLE
V12 | Driver: Ravi | Type: Compactor | Capacity: 2000kg | AVAILABLE
V13 | Driver: Manoj | Type: Heavy Truck | Capacity: 5000kg | AVAILABLE
=====
SMART MUNICIPAL WASTE MANAGEMENT
=====
1. Register Resident
2. Register Collection Vehicle
3. Show Bin Fill Status
4. Assign Collection Request
5. Cancel Collection Request
6. Search Resident
7. Route Efficiency Report
8. Run Bin & Route Monitoring
9. View Alerts
```

```
=====
SMART MUNICIPAL WASTE MANAGEMENT
=====
1. Register Resident
2. Register Collection Vehicle
3. Show Bin Fill Status
4. Assign Collection Request
5. Cancel Collection Request
6. Search Resident
7. Route Efficiency Report
8. Run Bin & Route Monitoring
9. View Alerts
10. View Waitlist
11. View Vehicles
12. View Collection Requests
0. Exit
=====
Enter your choice: 0

System closed safely.

...Program finished with exit code 0
Press ENTER to exit console.
```